

LEARNING TARGET

Distinguish qualitative and quantitative observations, choose an appropriate tool, and record a complete SI-style measurement.

NOTICE / PREDICT

Two students describe the same sample as 'heavy' and '18.62 g.' Which record can another scientist compare directly, and what makes it more useful?

Be ready to defend your first idea. You may revise it after the lesson.



Electronic balance: mass

Explain the idea

BIG IDEA

A measurement is evidence only when the value, unit, quantity, tool, and reasonable precision are communicated clearly.

ESSENTIAL TERMS

qualitative: descriptive evidence without a measured number

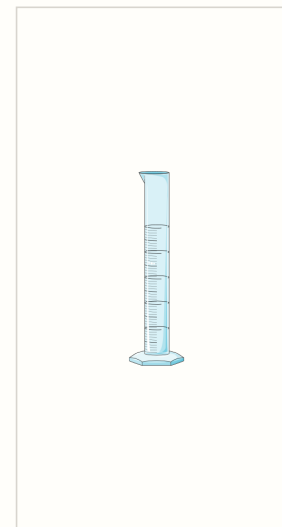
quantitative: evidence that includes a measured or counted number

resolution: the smallest increment an instrument can display or support

SI unit: a standard unit used to communicate a physical quantity



Electronic balance: mass



Graduated cylinder: liquid volume



Beaker: holding and approximate volume

WHAT TO REMEMBER

- A complete quantitative record names the measured quantity and gives a number with a unit.
- Choose the tool for the quantity and needed resolution: a balance measures mass; a graduated cylinder measures liquid volume more precisely than a beaker.
- Write a space between a value and its unit symbol, use correct capitalization, and do not pluralize symbols: 6.0 mL, not 6.0mls.
- A leading zero prevents a missed decimal: write 0.45 g, not .45 g.

Model the reasoning

WORKED EXAMPLE

A data table contains only '42.7.' Diagnose the record and repair it for a liquid-volume measurement.

- 1 Identify what is missing: the record does not name the measured quantity or unit.
- 2 Use the measurement context to choose the correct unit: milliliters for this liquid volume.
- 3 Preserve the reported precision and apply SI spacing and capitalization.

RESULT

Record the result as 'liquid volume = 42.7 mL' and identify the instrument used.

CHECK: Why is 24.6 mL more informative than the number 24.6 alone?

LEARNING TARGET

Represent measured values in scientific notation and apply significant-figure rules to calculations.

NOTICE / PREDICT

Two calculators display 39.68 and 39.680000. Does the longer display automatically make the experimental answer more precise?

Be ready to defend your first idea. You may revise it after the lesson.

Explain the idea

BIG IDEA

Scientific notation preserves place value; significant figures communicate justified precision from measured data.

ESSENTIAL TERMS

coefficient: the number from 1 up to, but not including, 10 in scientific notation

exponent: the power of ten that records decimal movement

significant figure: a digit justified by the measurement and its recorded precision

exact number: a counted or defined value that does not limit significant figures

WHAT TO REMEMBER

- Scientific notation has the form $a \times 10^n$, where $1 \leq a < 10$.
- Leading zeros are placeholders; zeros between nonzero digits and recorded trailing decimal zeros are significant.
- For multiplication and division, round to the fewest significant figures among measured values.
- For addition and subtraction, round to the least precise decimal place. Keep guard digits and round once at the end.

WORKED EXAMPLE

Calculate 12.4×3.20 and report the answer with justified precision.

1

Calculate without early rounding: $12.4 \times 3.20 = 39.68$.

2

Count significant figures: 12.4 has 3 and 3.20 has 3.

3

Round the product to 3 significant figures.

RESULT**39.7**

CHECK: Why is 39.68 not the final reported answer for the worked example?

LEARNING TARGET

Convert quantities by arranging conversion factors so unwanted units cancel and the target unit remains.

NOTICE / PREDICT

**A conversion gives the expected number,
but the unwanted unit never cancels.
Should the result be trusted?**

Be ready to defend your first idea. You may revise it after the lesson.



A rate such as km/h combines two units; both must be tracked during conversion.

BIG IDEA

A conversion factor equals one. Its orientation is chosen by units, and unit cancellation is the built-in reasonableness check.

ESSENTIAL TERMS

conversion factor: a ratio of equivalent quantities that equals one

factor-label method: a method that tracks and cancels units through multiplication

target unit: the unit required in the final answer

dimensional consistency: agreement between the units in a setup and the intended quantity



A rate such as km/h combines two units; both must be tracked during conversion.

WHAT TO REMEMBER

- Start with the given quantity and write its unit.
- Place each unwanted unit opposite itself so it cancels.
- Continue multiplying by equal ratios until only the target unit remains.
- Defined conversion factors, such as $1000\text{ m} = 1\text{ km}$, are exact and do not limit significant figures.

WORKED EXAMPLE

Convert 72.0 km/h to m/s.

1

Write $72.0 \text{ km/h} \times (1000 \text{ m} / 1 \text{ km})$ so kilometers cancel.

2

Multiply by $(1 \text{ h} / 3600 \text{ s})$ so hours cancel.

3

Calculate $72.0 \times 1000 / 3600$ and retain three significant figures.

RESULT**20.0 m/s**

CHECK: What does it mean if both kilometers and meters remain in the final unit?

LEARNING TARGET

Calculate density, rearrange the density relationship, and use percent error to compare an experimental result with an accepted value.

NOTICE / PREDICT

Two objects have different masses but the same density. What does density reveal that mass alone does not?

Be ready to defend your first idea. You may revise it after the lesson.



Read liquid volume at eye level; use the bottom of a concave meniscus.

BIG IDEA

Density relates mass to volume, while percent error measures the size of a difference relative to an accepted value.

ESSENTIAL TERMS

density: mass per unit volume, $d = m / V$

accepted value: the reference value used for comparison

experimental value: the value produced by a measurement or calculation

percent error: absolute difference divided by accepted value, multiplied by 100



Read liquid volume at eye level; use the bottom of a concave meniscus.

WHAT TO REMEMBER

- Density units combine mass and volume, commonly g/mL or g/cm³.
- Use $d = m / V$; rearrange to $m = dV$ or $V = m/d$ when needed.
- Percent error = $| \text{experimental} - \text{accepted} | / \text{accepted} \times 100\%$.
- Keep the absolute value in the numerator and use the accepted value as the denominator.

WORKED EXAMPLE

A sample has a mass of 57.0 g and a volume of 20.0 mL. Calculate its density.

1

Select $d = m / V$ and substitute measured values with units.

2

Calculate $57.0 \text{ g} / 20.0 \text{ mL} = 2.85 \text{ g/mL}$.

3

Both measurements have three significant figures, so report three.

RESULT**2.85 g/mL**

CHECK: Why must a density answer include both a mass unit and a volume unit?

LEARNING TARGET

Use repeated measurements and an accepted value to distinguish precision, accuracy, random variation, and systematic error.

NOTICE / PREDICT

Three measurements agree closely with one another but are all higher than the accepted value. Is the method precise, accurate, both, or neither?

Be ready to defend your first idea. You may revise it after the lesson.



A zero offset can create a systematic mass error.

BIG IDEA

Precision is agreement among repeated trials; accuracy is agreement with an accepted value. One statistic cannot establish both.

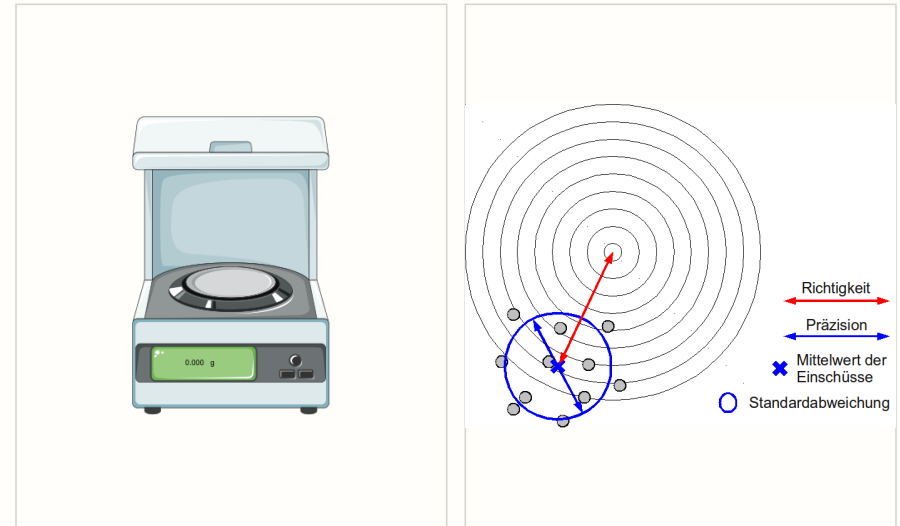
ESSENTIAL TERMS

precision: closeness of repeated measurements to one another

accuracy: closeness of a result to an accepted value

random error: unpredictable variation that changes from trial to trial

systematic error: a consistent offset caused by method or instrument bias



A zero offset can create a systematic mass error.

Center agreement and trial spread answer different questions.

WHAT TO REMEMBER

- Judge precision from the spread of repeated measurements, not from one trial.
- Judge accuracy by comparing a result or mean with an accepted value.
- Random variation increases scatter; repeated trials reveal its size.
- Systematic error shifts results in a consistent direction and is not fixed by averaging more biased trials.

WORKED EXAMPLE

Trials are 1.247, 1.249, and 1.248 g/mL; the accepted value is 1.200 g/mL. Evaluate the results.

1

Compare trials: the range is only 0.002 g/mL, so the measurements are precise.

2

Find the mean: $(1.247 + 1.249 + 1.248) / 3 = 1.248$ g/mL.

3

Compare with 1.200 g/mL: the mean is consistently high; percent error is 4.00%.

RESULT

Precise but not accurate relative to the accepted value; the pattern suggests a systematic high bias.

CHECK: Why would ten more trials not automatically remove a balance zero offset?